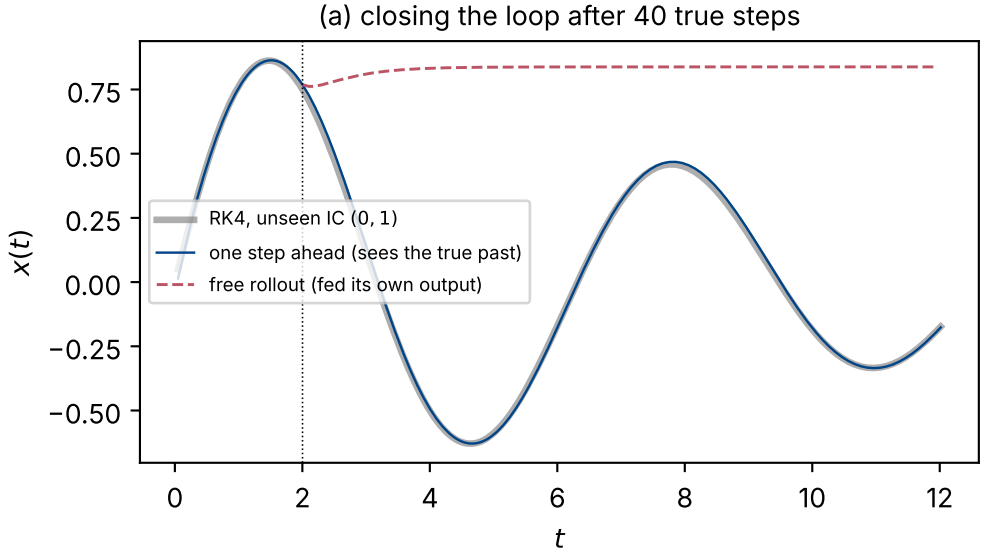
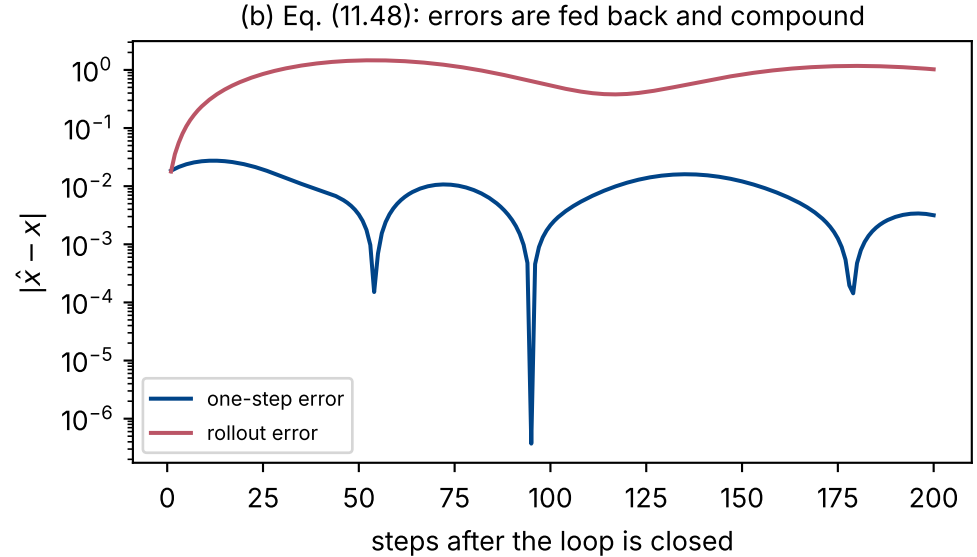

(a) closing the loop after 40 true steps
RK4, unseen IC (0, 1)
one step ahead (sees the true past)
free rollout (fed its own output)
$x(t)$
t
0.75
0.50
0.25
0.00
−0.25
−0.50
0
2
4
6
8
10
12


(b) Eq. (11.48): errors are fed back and compound
one-step error
rollout error
$|\hat{x} - x|$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
25
50
75
100
125
150
175
200
steps after the loop is closed